Quadratic Forms and Genus Theory, and $p = x^2 + ny^2$

Yuval Amit

Quadratic Forms

Definition

A **quadratic from in two variables**, which we'll refer to as a "*form*", is an expression $ax^2 + bxy + cy^2$, with $a, b, c \in \mathbb{Z}$. We say this form is **primitive** if $\gcd(a, b, c) = 1$.

An integer m is **represented** by a form f if $f(x, y) = m$ has a solution. m is **properly represented** if $\gcd(x, y) = 1$.

Definition

Two forms f, g are **equivalent** if $f(x, y) = g(mx + ny, rx + sy)$ for some $m, n, r, s \in \mathbb{Z}$, with $ms - nr = \pm 1$. They are **properly equivalent** if $ms - nr = 1$. Both regular and proper equivalence define an equivalence relation.

The Discriminant

Definition

The **discriminant** of a form $f(x, y) = ax^2 + bxy + cy^2$ is $b^2 - 4ac$.

Proposition

Let f and g be equivalent forms. Then,

1. f and g represent (and properly represent) the same numbers
2. f and g have the same discriminant.

Proposition

If f has a negative discriminant, then either f represents only positive integers, in which case we say f is **positive definite**, or f represents only negative integers, in which case we say f is **negative definite.**

Reduced Forms

Definition

A primitive (coprime coefficients) positive definite form is **reduced** if

$$|b| \leq a \leq c, \text{and if } (a = |b| \text{ or } a = c), \text{ then } b \geq 0$$

Do not think about this definition for more than two seconds.

Theorem

Let f be primitive and positive definite with discriminant D. Then,

1. f is properly equivalent to a unique reduced form with discriminant D.
2. There are finitely many reduced forms with discriminant D.

This means we have a finite number equivalence classes (with proper equivalence) of primitive pos. def. forms with discriminant D.

The Class Group

Definition

For a fixed discriminant D, let $Cl(D)$ be this finite set of equivalence classes, called the **class group** (spoilers: it's a group). The size of the class group, $h(D)$, is called the **class number.**

Lemma

Let $D \equiv 0 \pmod 4$ be nonzero. There exists a unique homomorphism $\chi : (\mathbb{Z}/D\mathbb{Z})^\times \to \{\pm 1\}$ such that $\chi(p) = \left(\frac{D}{p}\right)$ for an odd prime $p \nmid D$.

Theorem (!!!)

Let $D \equiv 0 \pmod 4$, and $p \nmid D$ an odd prime. Then, p is represented by one of the $h(D)$ reduced forms with discriminant D if and only if $p \in \ker \chi \iff \left(\frac{D}{p}\right) = 1$.

Uses and Limitations

Example

We can check that $\left(\frac{-4}{p}\right) = 1 \iff p \equiv 1 \pmod 4$. Thus, any odd prime $p \equiv 1 \pmod 4$ is properly represented by a reduced form with discriminant -4. But, one can very the only such reduced form is $x^2 + y^2$. This proves $p = x^2 + y^2 \iff p \equiv 1 \pmod 4$.

Example

By similar methods, a prime p is represented by a reduced form with discriminant $-20 \iff p \equiv 1, 3, 7, 9 \pmod{20}$. However, there are two reduced forms with this discriminant: $x^2 + 5y^2$ and $2x^2 + 2xy + 3y^2$. So, we don't have a complete answer for $p = x^2 + 5y^2$

$x^2 + ny^2$ is always reduced, and has discriminant $-4n$. So, we can get a complete solution for $p = x^2 + ny^2$ when $h(-4n) = 1$. Landau proved this only happens for $n = 1, 2, 3, 4, 7$

What Now?

Observation

Recall the reduced forms of discriminant -20:

$$x^2 + 5y^2, 2x^2 + 2xy + 3y^2$$

and they represent primes $p \equiv 1, 3, 7, 9 \pmod{20}$. Lagrange found $x^2 + 5y^2$ never represents 3 or 7 mod 20, while $2x^2 + 2xy + 3y^2$ never represents 1 or 9 mod 20. Thus, we can confidently say

$$p = x^2 + 5y^2 \iff p \equiv 1, 9 \pmod{20}$$

We can learn more about the reduced forms in the class group by studying the values they represent.

The Genus

Definition

Two primitive, positive definite forms with discriminant D are in the same **genus** if they represent the same values in $(\mathbb{Z}/D\mathbb{Z})^\times$

Remark

Equivalent forms represent the same values, hence are in the same genus.

The Principal Form

Definition

Let $D \equiv 0 \pmod 4$ be negative. The **principal form** in $Cl(D)$ is $x^2 - \frac{D}{4}y^2$. We call the genus containing this form the **principal genus**.

Remark

$x^2 + ny^2$ is the principal form in $Cl(-4n)$ for $n \geq 1$.

Theorem

Let $D \equiv 0 \pmod 4$ be negative.

1. The values represented in the principal genus form a subgroup $R_{prin} \leq \ker \chi$
2. For any f with discriminant D, the values R_f in $(\mathbb{Z}/D\mathbb{Z})^\times$ represented by f form a coset of R_{prin}.

The Class Group is a Group

Remark

Let $D \equiv 0 \pmod 4$ be negative. There is a well defined abelian group operation on the classes of forms in $Cl(D)$, that in particular satisfies:

1. The class of the principal form (the **principal class**) is the identity.
2. If f represents m, and g represents n, then $[f] + [g]$ represents mn.

Lemma

Let $\Phi : Cl(D) \to \ker(\chi)/R_{prin}$ be the map which sends a class to the coset of R_{prin} that it represents, i.e. its "genus". Then Φ is a group homomorphism.

Corollary

All genera of forms of discriminant D consist of the same number of classes.

Almost Done

Since the values represented by distinct genera are completely disjoint (they're cosets), we can refine the relation between $p = x^2 + ny^2$ and forms.

Theorem

Let $D \equiv 0 \pmod 4$ be negative. If R_f is a coset of R_{prin} in $\ker \chi$, and $p \nmid D$ is prime, then $p \in R_f \iff p$ is represented by a reduced form of discriminant D in the genus corresponding to R_f.

Remark

$x^2 + ny^2 \equiv x^2, x^2 + n \pmod{4n}$. Thus, p is represented by a form in the principal genus if and only if $p \equiv \beta^2, \beta^2 + n$ for some integer β.

Corollary

If the size of the genera for a given discriminant is 1, then we have an exact congruence condition for $p = x^2 + ny^2$.

Where Does This Get Us?

This raises the question: when does a class group have genus size one?

Theorem

Let $D \equiv 0 \pmod 4$ be negative. Then, the genera in $Cl(D)$ have size 1 if and only if the class number $h(-4n) = 2^{\mu-1}$, where μ is the number of prime divisors of $-4n$.

$h(-4n)$	n's with one class per genus
1	1,2,3,4,7
2	5,6,8,9,10,12,13,15,16,18,22,25,28,37,58
4	21,24,30,33,40,42,45,48,57,60,70,72,78,85,88,93,102,112 130,133,177,190,232,253
8	105,120,165,168,210,240,273,280,312,330,345,357,385 408,462,520,760
16	840,1320,1365,1848